Jacob Webster

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Education

University of South Carolina — Columbia, SC
Bachelor of Science in Computer Science | GPA: 3.6 — *Aug 2024 – May 2027*

- **Relevant Coursework:** C++ Programming, Data Structures & Algorithms, Machine Learning Systems, Operating Systems, Computer Networks, Software Engineering, Statistics for Engineers

Experience

Software Engineer (Part-Time) — Sep 2026 – Present
Michelin — *Greenville, SC*

- Revamp and update technical documentation for the team's applications to support onboarding
- Contribute to the team's testing and code quality tooling, including Renovate dependency updates, SonarQube static analysis, and automated test setup

Software Engineering Intern — May 2026 – Aug 2026
Michelin — *Greenville, SC*

- Developed automated frontend tests using Playwright for Michelin's inventory management application, decreasing minor release testing time from 150 minutes to 3.5 minutes
- Optimized data pipelines by reducing steps to transmit parcel data by 75% using Kafka and Java microservices

Software Engineering Fellow — Aug 2025 – Present
University of South Carolina – Center for Industry Solutions — *Columbia, SC*

- Developing a voice-commanded person-following system for Boston Dynamics Spot: speech-to-query parsing, VLM grounding (LocateAnything-3B), and YOLO11n tracking; integrating with Spot navigation
- Led a 3-person software team building Caelus, a ROS2 drone perception prototype that accumulates 2D LiDAR scans into a 3D voxel point cloud (10 cm resolution); set Git workflow and code review standards
- Built an ESP32 relay (921,600 baud) to replace a failing UART adapter, restoring reliable LiDAR data into ROS2

Projects

WayFinder | *Python, LightGBM, Pandas* | ShellHacks X — Sep 2026

- Built a LightGBM model that infers traffic light state from nearby vehicle behavior, giving autonomous vehicles a fallback during signal outages
- Hit 95% accuracy when confident and cut false-green errors to 2.5% (vs. 19% baseline), at 0.25 ms per inference
- Engineered per-lane features from 2M+ rows across 16k Waymo scenarios, with ego-lane detection and leakage controls

Baton | *Python, FastAPI, React, WebSockets, SQLite* | PalmettoHacks — Oct 2026

- Contributed to a multi-user layer for AI coding agents, streaming every agent step to all session participants over WebSockets, with roles, hand-offs, and a replayable timeline
- Designed an append-only event log (SQLite WAL) that rebuilds agent context each turn, enabling crash recovery, session forking, and switching between Claude, Gemini, OpenAI, and local models mid-session

Technical Skills

Languages: Python, C++, Java, C, JavaScript/TypeScript, SQL
Robotics & ML: ROS2, LiDAR/point clouds, LightGBM, NumPy, Pandas, PyTorch, TensorFlow, YOLO, VLMs
Systems & Backend: Linux, Git, Kafka, Spring Boot, REST APIs, FastAPI, WebSockets, SQLite
Web & Testing: React, Node.js, TailwindCSS, Playwright

Honors & Awards

Scholarships & Academic: Palmetto Fellows Scholarship (SC merit-based), Dean's Scholar (5 semesters)
Hackathons: CUhackit 2025, Best Web Dev; NASA Space Apps, Most Inspirational; PalmettoHacks, 2nd Overall